Chapter 8: Understanding t-test Output

Interpreting Group Comparisons

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1 Overview

This reading prepares you for the lab session where you'll interpret **t-test** output. The t-test is one of the most widely used statistical tests in psychology, allowing us to compare means and determine whether observed differences are likely to reflect real effects or just random variation.

> **ℹ What You'll Learn**
>
> By the end of this lab, you'll be able to:
>
> - Interpret t-test output including t-values, p-values, and confidence intervals
> - Understand the difference between independent and paired samples t-tests
> - Choose the appropriate type of t-test for your research question
> - Report t-test results in APA format

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2 When to Use a t-test

The t-test answers the question: **Is the difference between two means larger than we'd expect by chance?**

Use a t-test when you have:

1. A **continuous** outcome variable (e.g., reaction time, mood score)
2. **Two groups** to compare OR two measurements from the same people
3. Reasonably **normally distributed** data (or a large enough sample)

2.1 Types of t-tests

Type	When to Use	Example
Independent samples	Different people in each group	Room A vs Room B
Paired samples	Same people measured twice	Pre vs Post mood
One sample	Compare to a known value	Sample mean vs 50%

3 Independent Samples t-test

Use this when comparing **different people** in two groups.

3.1 Example: Comparing Rooms

Did Room A and Room B differ in reaction time?

3.2 Understanding the Output

```
    Welch Two Sample t-test

    data:  reaction_time by room
    t = -2.34, df = 45.2, p-value = 0.024
    alternative hypothesis: true difference in means is not equal to 0
    95 percent confidence interval:
     -42.5  -3.2
    sample estimates:
    mean in group A   mean in group B
              265.3             288.1
```

3.3 What Each Part Means

Output	Meaning
`t = -2.34`	The t-statistic: how many standard errors the difference is from 0
`df = 45.2`	Degrees of freedom
`p-value = 0.024`	Probability of this difference (or larger) if null hypothesis is true
`95 percent confidence interval`	Range of plausible values for the true difference
`sample estimates`	The actual group means

3.4 Key Questions to Answer

1. **What are the means?** → Room A: 265.3 ms, Room B: 288.1 ms
2. **What is t?** → -2.34
3. **Is it significant?** → Yes, $p = 0.024 < 0.05$
4. **What's the CI?** → $[-42.5, -3.2]$ - the true difference is likely in this range
5. **What does it mean?** → Room A was faster than Room B

! Interpreting the p-value

A p-value of 0.024 means: **if there were truly no difference between rooms**, we'd see a difference this large (or larger) about 2.4% of the time. Since this is less than 0.05, we conclude the difference is statistically significant.

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4 Paired Samples t-test

Use this when comparing **the same people** measured at two time points.

4.1 Example: Pre-Post Mood

Did mood change after DataLab?

4.2 Understanding the Output

```
    Paired t-test

    data:  mood_pre and mood_post
    t = -3.45, df = 29, p-value = 0.002
    alternative hypothesis: true mean difference is not equal to 0
    95 percent confidence interval:
     -1.82  -0.45
    sample estimates:
    mean difference
              -1.13
```

4.3 Key Questions to Answer

1. **What is the mean difference?** → −1.13 (mood increased by 1.13 points)
2. **Is it significant?** → Yes, $p = 0.002 < 0.05$
3. **What's the CI?** → [−1.82, −0.45] - the true change is likely in this range
4. **What does it mean?** → Mood significantly improved after DataLab

4.4 Why Paired Tests Are More Powerful

Paired tests account for individual differences. Consider:

- Person A: Pre = 5, Post = 7 (change = +2)
- Person B: Pre = 8, Post = 10 (change = +2)

Both people increased by 2 points, even though their baseline scores differ. The paired test focuses on the **change within each person**, not the raw scores.

Rule of Thumb

If the same individuals provide both measurements, use a paired test. It's more sensitive because it removes between-person variability.

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5 One Sample t-test

Use this to compare your sample mean to a specific value.

5.1 Example: Comparing to Chance

Is staring detection accuracy different from 50% (chance)?

5.2 Understanding the Output

```
    One Sample t-test

data:  staring_accuracy
t = 1.87, df = 29, p-value = 0.072
alternative hypothesis: true mean is not equal to 0.5
95 percent confidence interval:
 0.48  0.61
sample estimates:
mean of x
    0.545
```

5.3 Key Questions to Answer

1. **What is the sample mean?** → 54.5%
2. **Is it significantly different from 50%?** → No, $p = 0.072 > 0.05$
3. **What does it mean?** → Performance is not significantly better than chance

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6 Effect Size: Cohen's d

The p-value tells you whether a difference is **statistically significant**, but not whether it's **practically meaningful**. For that, we use effect sizes.

Cohen's d expresses the difference in standard deviation units:

d value	Interpretation
0.2	Small effect
0.5	Medium effect
0.8	Large effect

6.1 Example Output

```
Cohen's d

d estimate: 0.62
95% CI: [0.15, 1.09]

Interpretation: medium
```

> **! Why Effect Size Matters**
>
> A study with n = 1000 might find p < .001 for a tiny effect (d = 0.1). The effect is "real" but practically meaningless. Always report and interpret effect sizes alongside p-values.

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7 Choosing the Right Test

Research Question	Design	Test
Do two different groups differ?	Between-subjects	Independent t-test
Does the same group change over time?	Within-subjects	Paired t-test
Does a sample differ from a known value?	One-sample	One-sample t-test

7.1 DataLab Examples

Question	Test Type
Did Room A and B differ on reaction time?	Independent
Did mood change from pre to post?	Paired
Was throwing accuracy above 50%?	One-sample
Did word type (abstract/concrete) affect memory?	Independent

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8 Assumptions of the t-test

8.1 1. Independence

Observations should be independent (one person's score doesn't influence another's). This is about study design.

8.2 2. Normality

The data should be approximately normally distributed, especially for small samples.

> Robustness
>
> The t-test is quite robust to violations of normality, especially with larger samples ($n > 30$ per group). Don't worry too much unless your data are severely skewed.

8.3 3. Homogeneity of Variance (for independent samples)

Groups should have similar variability. The Welch's t-test (default in most software) handles unequal variances automatically.

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9 Reporting t-tests (APA Style)

9.1 Template

> [Group 1] (M = XX.X, SD = X.X) [were/was] [significantly/not significantly] [higher/lower/different] than [Group 2] (M = XX.X, SD = X.X), t(df) = X.XX, p = .XXX, d = X.XX.

9.2 Independent Samples Example

> Participants in Room A ($M = 265.3$ ms, $SD = 45.2$) had significantly faster reaction times than those in Room B ($M = 288.1$ ms, $SD = 52.1$), $t(45.2) = -2.34$, $p = .024$, $d = 0.47$.

9.3 Paired Samples Example

> Mood significantly increased from pre-session ($M = 5.2$, $SD = 1.8$) to post-session ($M = 6.3$, $SD = 1.6$), $t(29) = 3.45$, $p = .002$, $d = 0.62$.

9.4 Elements to Include

- Group means and SDs
- The t-value and df
- The p-value
- Effect size (Cohen's d)
- Direction of the difference

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10 Common Interpretation Errors

10.1 Error 1: "$p > .05$ means no difference"

Wrong! A non-significant result means we can't rule out chance — not that there's definitely no effect. The study might lack power.

10.2 Error 2: "p < .001 means a huge effect"

Wrong! Statistical significance depends on sample size. Large samples can detect tiny effects. Always check effect size.

10.3 Error 3: "t = 2.34 is bigger than t = 1.98, so it's more significant"

Misleading! t-values depend on sample size and variability. Compare p-values and effect sizes instead.

10.4 Error 4: Confusing paired and independent tests

Check: Are the same people in both conditions? If yes → paired. If no → independent.

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11 Preparation Checklist

Before arriving at the lab:

> ⚠ Before You Arrive
>
> - ☐ Read this chapter carefully
> - ☐ Review the lecture material on hypothesis testing
> - ☐ Understand the difference between paired and independent samples
> - ☐ Be able to interpret p-values correctly
> - ☐ Know what Cohen's d means

11.1 Questions to Consider

- What research questions from DataLab can we answer with t-tests?
- Which comparisons require paired tests vs independent tests?
- What would a significant result mean in practical terms?

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12 Summary

Concept	Key Point
t-test purpose	Compare means to see if difference is beyond chance
Independent samples	Different people in each group
Paired samples	Same people measured twice
p-value	Probability of data if null hypothesis is true
Effect size (d)	Practical magnitude of the difference
APA format	Include M, SD, t, df, p, and d

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13 Quick Reference: Interpreting Output

13.1 Independent Samples

Look for	Tells you
Group means	Which group scored higher
t-value	Direction and magnitude of difference
p-value	Whether difference is significant
CI	Range of plausible true differences
d	Practical size of effect

13.2 Paired Samples

Look for	Tells you
Mean difference	Average change
t-value	Direction and magnitude of change
p-value	Whether change is significant
CI	Range of plausible true change
d	Practical size of effect